Forward- and Backward-Looking Dependency Processing in L1 and L2 English

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Introduction: In everyday communication, comprehenders must both predict upcoming material and retrieve earlier information from memory. Forward-looking (predictive) and backward-looking (integrative) dependencies provide a useful testbed for how L1 and L2 speakers coordinate these processes over time. Prior work suggests that L2 speakers may rely less on detailed syntactic structure and may be especially challenged by backward-looking dependencies that depend on memory retrieval rather than forward prediction (Clahsen & Felser, 2018; Cunnings, 2017; Felser, 2019; Grüter et al., 2017). However, few studies have directly compared forward vs. backward dependencies across L1 and L2 groups using parallel materials while also modeling individual differences in cognitive and linguistic abilities. This study asks (i) how L1 English and L1-Farsi L2 English speakers process forward- and backward-looking pronominal dependencies with gender match vs. mismatch (4 conditions, Table 1), (ii) how online results relate to offline interpretation, and (iii) how working memory and lexical proficiency modulate processing.

Method: 50 L1 and 50 L2 English speakers are being recruited; data collection is ongoing. The pilot results reported here are based on 10 participants (5 L1, 5 L2). Participants completed the experiment online. The main task was a G-maze with 32 critical items adapted from Kennison et al. (2009) (Table 1), crossing Dependency Direction (anaphoric vs. cataphoric) with Gender Congruence (match vs. mismatch), plus 30 fillers. Word-by-word RTs were recorded. After each correctly completed sentence, a forced-choice comprehension question probed pronoun interpretation to link online processing with offline interpretation. Participants also completed the Extended English LexTALE (Fiorentino et al., 2024)and a backward digit span (BDS) task. Data were analyzed using linear and logistic mixed-effects models.

Preliminary Results: *Reaction Time*: Across conditions, L2 speakers showed slower RTs than L1 speakers. Log RTs at the pronoun/name region were analyzed with a linear mixed-effects model (Type III tests). In the L1 group there were significant effects of Dependency Direction, $F(1,1250.22)=7.69$, $p=.006$ (anaphora < cataphora), Gender Congruence, $F(1,778.07)=6.75$, $p=.010$ (match < mismatch), and a Direction × Congruence interaction, $F(1,592.07)=7.96$, $p=.005$, with a larger gender-mismatch penalty in anaphora (Figure1). For L2 speakers, the same model revealed no reliable main effects of Direction or Congruence and no interaction. *Coreference Interpretation*: Logistic mixed-effects modeling of comprehension questions showed a main effect of Group (L2 < L1; $\beta = -2.08$, $p=.012$) and a main effect of Gender Congruence (mismatch < match; $=\beta -1.96$, $p=.007$) (Figure 2). *Individual Differences*: A mixed-effects model including standardized BDS scores revealed that higher working-memory capacity predicted faster RTs across all conditions ($\beta=-0.16$, $p=.015$). In a parallel model, Extended LexTALE scores did not predict RTs for the current dataset.

Discussion: These preliminary findings suggest that L1 comprehenders show robust early sensitivity to both dependency direction and gender mismatch during pronoun resolution, whereas L2 comprehenders ultimately use gender information in offline judgments but show much weaker online effects (e.g.,Clahsen & Felser, 2006, 2018; Marinis et al., 2005). This pattern is consistent with slower or less automatic use of morpho-syntactic cues in the L2 and with possible cross-linguistic influence from Farsi, which is grammatically gender-neutral and lacks gender marking on third-person pronouns. The dissociation between online and offline measures in the L2 group thus aligns with accounts that propose shallower or delayed structural processing in a non-dominant language (Clahsen & Felser, 2006, 2018; Felser & Cunnings, 2012). The overall facilitatory effect of working memory further indicates that domain-general capacity supports efficient processing of both dependencies. Ongoing data collection, including larger samples and lexical-network measures, will allow a more fine-grained characterization of how dependency direction, language background, and individual differences jointly shape sentence processing. In addition, future work will test L2 speakers from L1 backgrounds with and without gender marking in pronouns to isolate possible effects of cross-linguistic influence.

Table 1 *Experimental conditions and example stimuli*

Condition	Example sentence
Anaphora_M/Mis	As **Ryan** approached the group, **he/she** shouted hello.
Cataphora_M/Mis	As **he/she** approached the group, **Ryan** shouted hello.

Figure 1 *Mean log reaction times in the region of interest (N–1, pronoun/name, N+1, N+2) for L1 English (left panel) and L2 English speakers (right panel) across all conditions.*

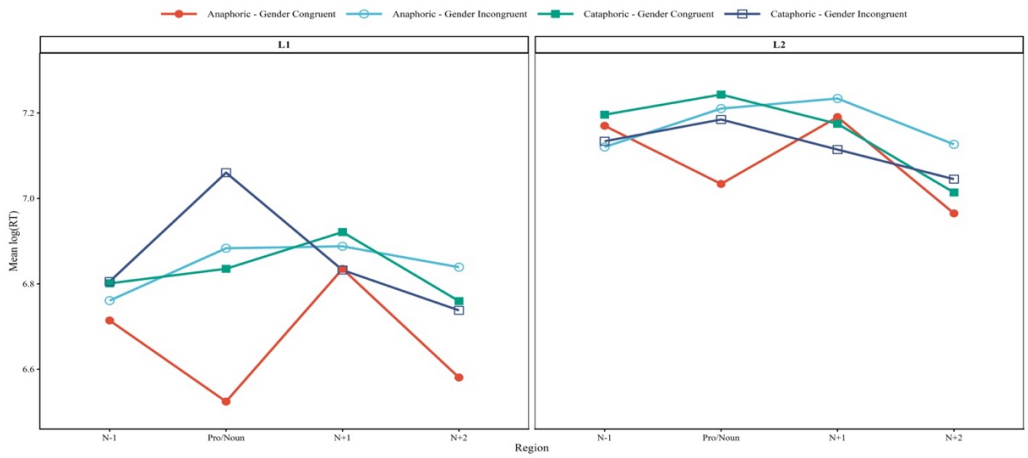
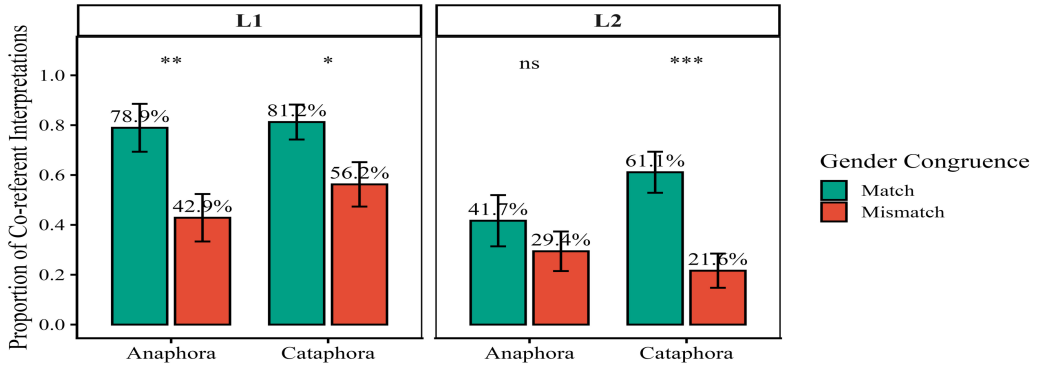


Figure 2 *Proportion of co-reference judgments as a function of dependency type (anaphora vs. cataphora), gender congruence (match vs. mismatch), and language group (L1 English) vs L2 English).*



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